

Theorem 5.2.1 Sum of the First n Integers

For every integer $n \geq 1$,

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

Proof (by mathematical induction): Let the property $P(n)$ be the equation

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Show that $P(1)$ is true:

To establish $P(1)$, we must show that

$$1 = \frac{1(1+1)}{2}$$

But the left-hand side of this equation is 1 and the right hand side is

$$\frac{1(1+1)}{2} = \frac{2}{2} = 1$$

also. Hence $P(1)$ is true.

Show that for every integer $k \geq 1$, if $P(k)$ is true then $P(k+1)$ is also true:

[Suppose that $P(k)$ is true for a particular but arbitrarily chosen integer $k \geq 1$. That is:] Suppose that k is any integer with $k \geq 1$ such that

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

[We must show that $P(k+1)$ is true. That is:] We must show that

$$1 + 2 + 3 + \dots + (k+1) = \frac{(k+1)(k+2)}{2}$$

[We will show that the left-hand side and the right-hand side of $P(k+1)$ are equal to the same quantity and thus are equal to each other.]

the left-hand side of $P(k+1)$ is

$$\begin{aligned}
& 1 + 2 + 3 + \dots + (k + 1) \\
&= 1 + 2 + 3 + \dots + k + (k + 1) \\
&= \frac{k(k + 1)}{2} + (k + 1) \\
&= \frac{k^2 + k}{2} + \frac{2k + 2}{2} \\
&= \frac{k^2 + 3k + 2}{2}
\end{aligned}$$

And the right-hand side of $P(k + 1)$ is

$$\frac{(k + 1)(k + 2)}{2} = \frac{k^2 + 3k + 2}{2}$$

This the two sides of $P(k + 1)$ are equal to the same quantity and so they are equal to each other. Therefore, the equation $P(k + 1)$ is true *[as was to be shown]*.

[Since we have proved both the basis step and the inductive step, we conclude that the theorem is true.]